

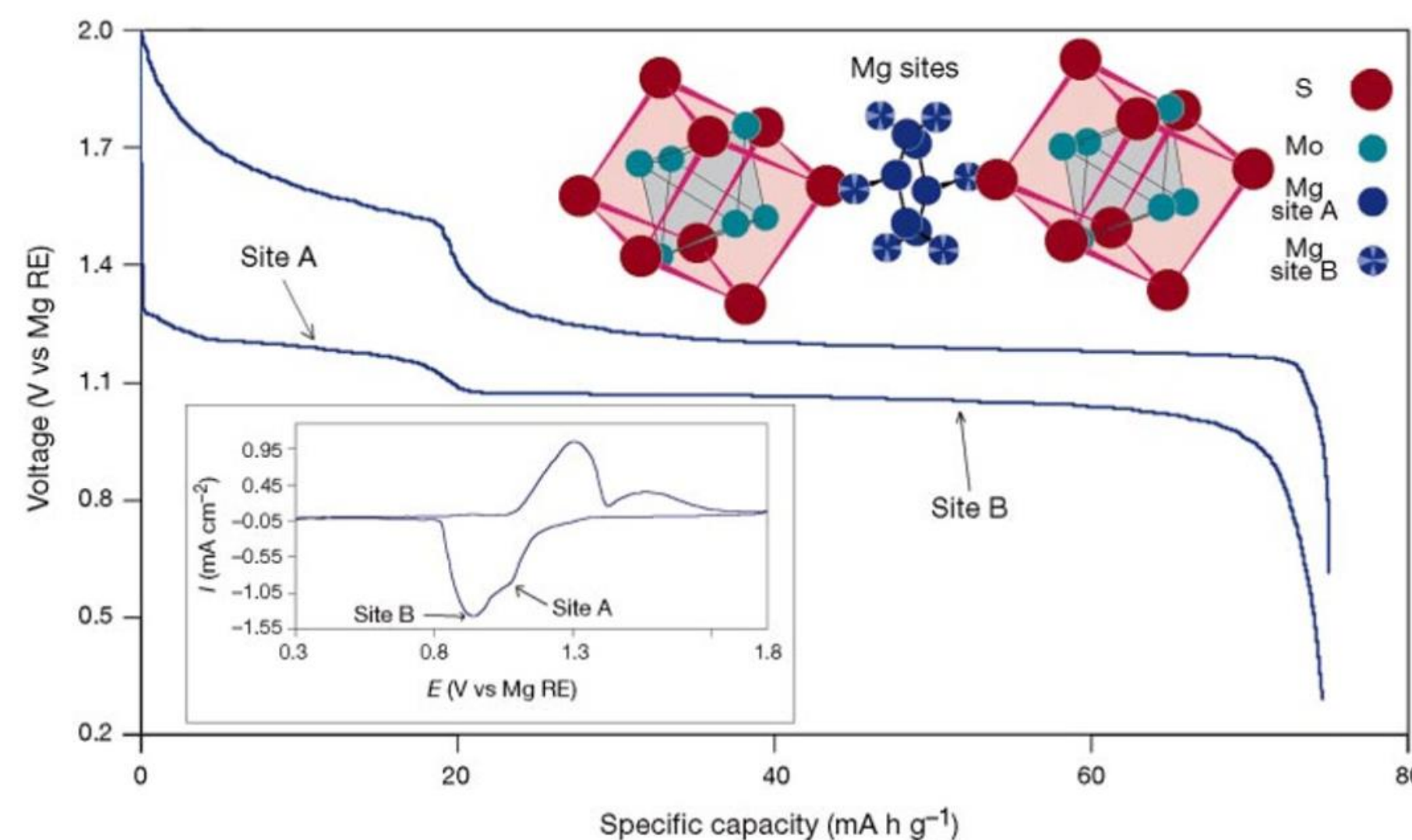
# A WEAKLY ION PAIRING ELECTROLYTE DESIGNED FOR HIGH VOLTAGE MAGNESIUM BATTERIES



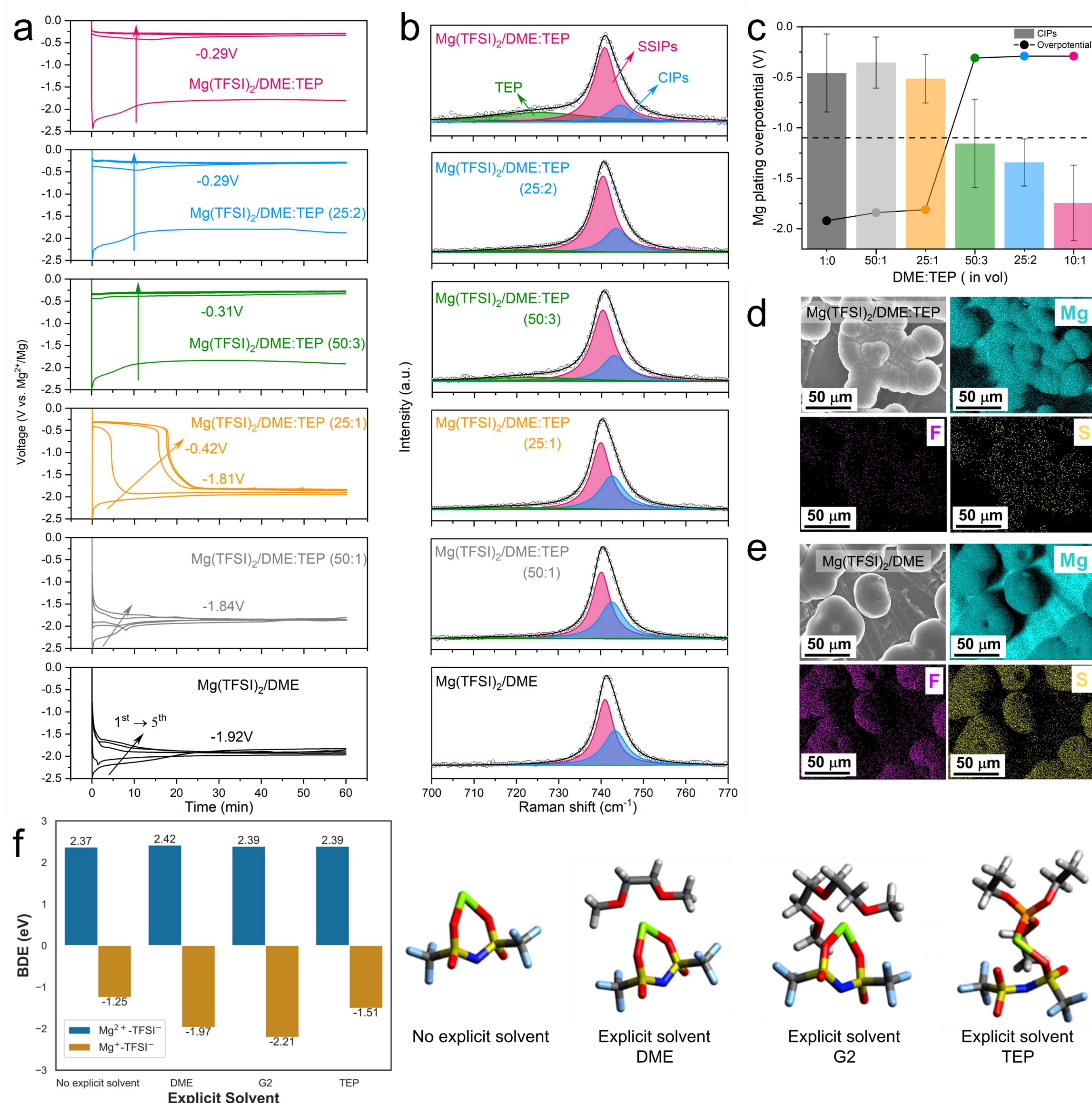
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## Motivation: develop high-voltage and non-corrosive Mg batteries

- The first-generation Mg battery ( $\text{Mo}_6\text{S}_8$  with chloride-based all-phenyl-complex (APC) electrolyte has high corrosivity, low output voltage and specific capacity [1]



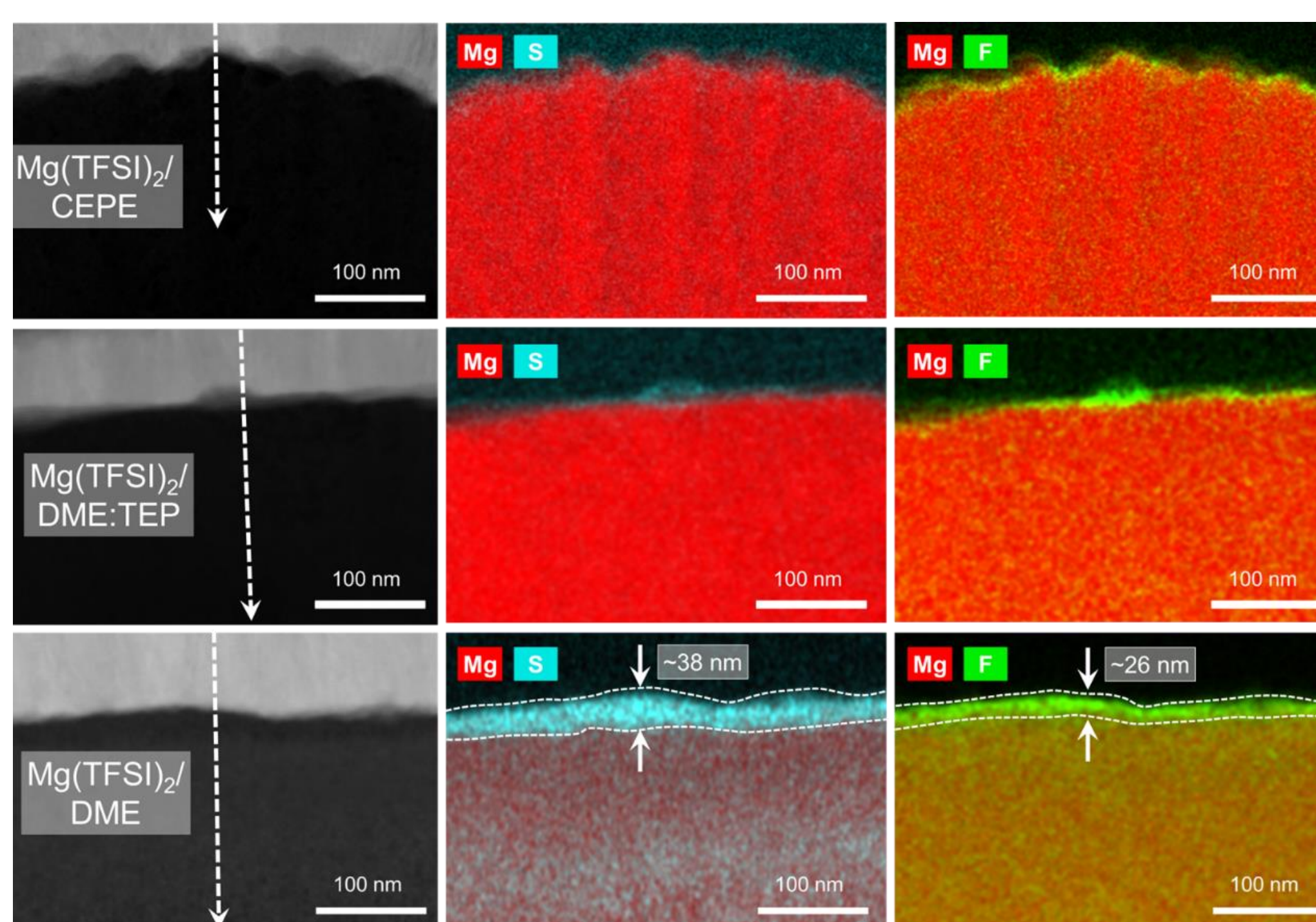
**Fig. 1 Tuning anion activities by controlling CIPs**



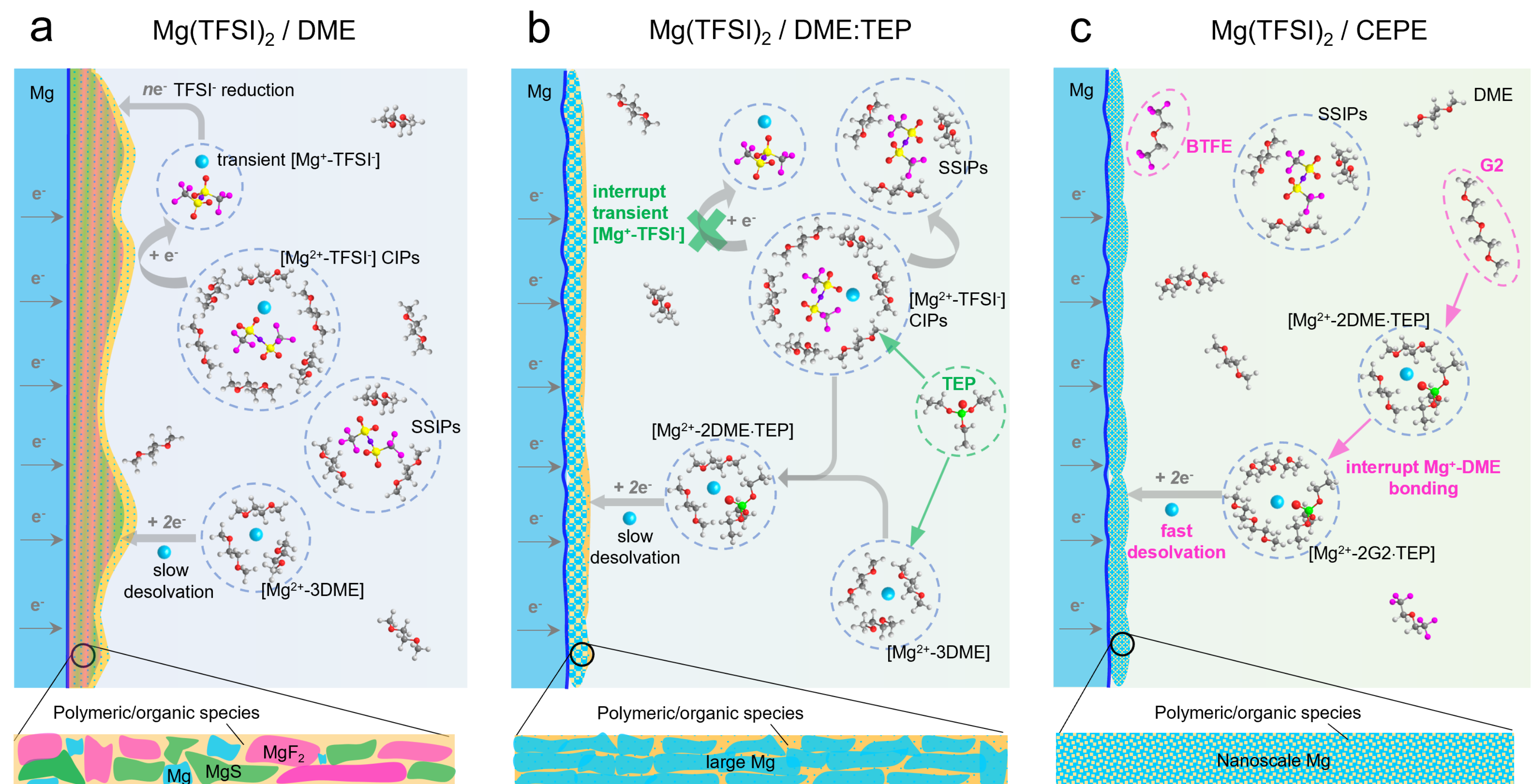
- Phosphate solvents effectively dissociate  $[\text{Mg}^{2+}\text{-TFSI}]$  CIPs
- The decomposition of  $[\text{Mg}^+\text{-TFSI}]$  CIPs was greatly suppressed
- Anion-derived interfacial F and S related greatly decreased

**Fig. 2 Interfacial SEI chemistry**

- TEM studies reveal the presence of thick passivation layer on deposited Mg in conventional electrolyte
- Transform thick anion-derived passivation layer into a thin  $\text{Mg}^{2+}$ -permeable SEI on Mg/electrolyte interface in CEPE



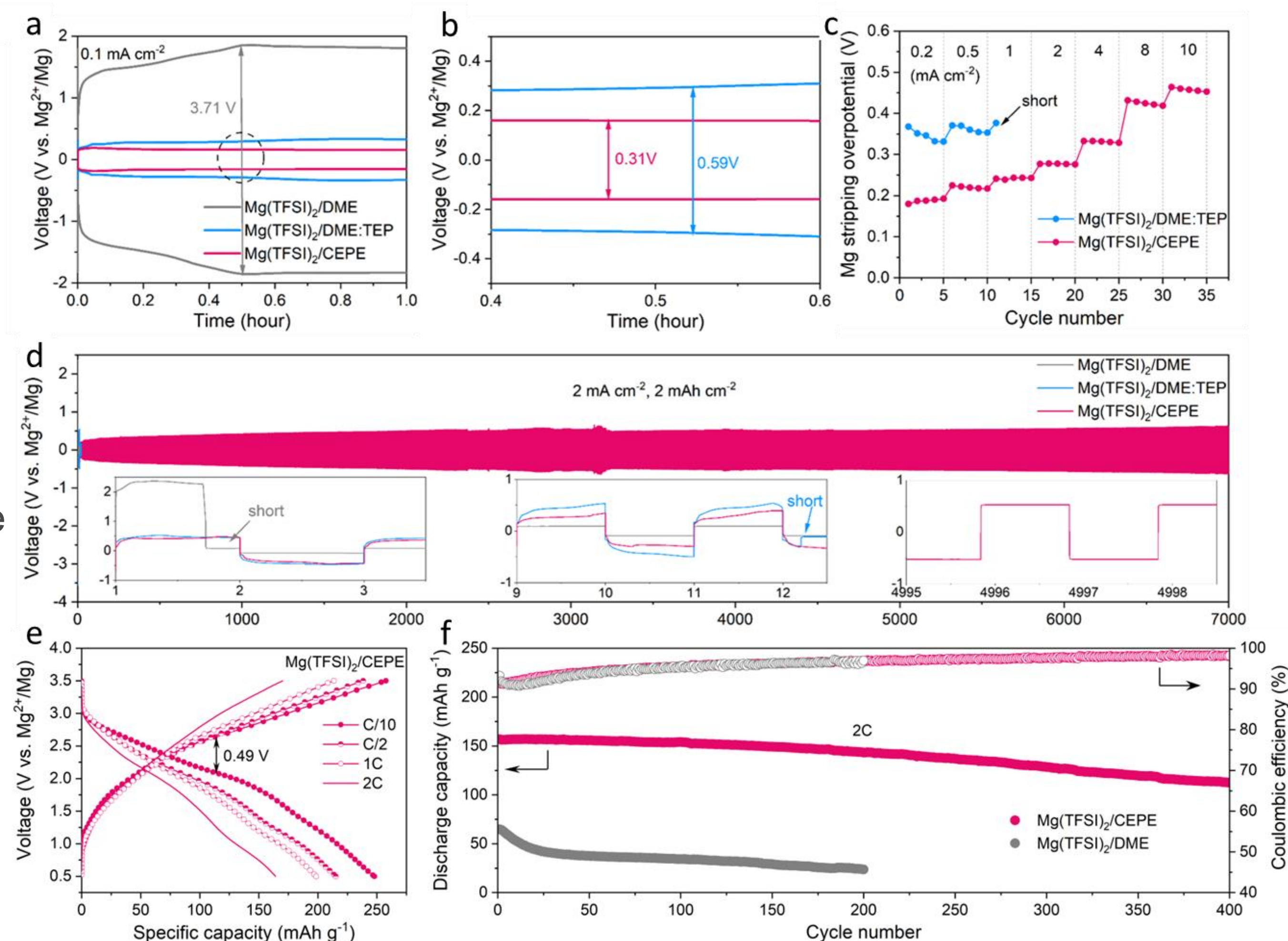
**Scheme 1. Electrolyte design strategy<sup>[2]</sup>**



- Solvent-designed strategy disrupts ion pairing without need for costly/bulky anion salts
- CEPE: Co-Ethereal Phosphate Electrolyte (solvent mixture of multiple ethers and phosphate)

**Fig. 3 Electrochemistry of symmetric cells and full cells**

- Highly reversible and stable Mg plating and stripping in CEPE
- Synergetic effect of phosphate and co-ethers
- Polyaniline (PANI) as anion-intercalative type cathode
- Suppressed CIP formation assists fast plating/stripping at the anode and facile TFSI<sup>-</sup> intercalation at the cathode



## Summary

- We report a new and easily accessible solvent-designed electrolyte, which effectively solves the difficulty of ion pair dissociation and facilitates fast nanoscale Mg nucleation/growth using simple  $\text{Mg}(\text{TFSI})_2$  as the salt, enabling a facile interfacial charge transfer process.
- The origin of this outstanding performance is the suppressed decomposition of contact-ion-pairs, obviating the formation of thick anion-derived passivation layer on Mg surface
- Full cell based on organic PANI cathode can work at an operating voltage up to 3.5 V that exhibit stable cycling at a 2C rate with 99% coulombic efficiency after 400 cycles.

## References

- [1] Aurbach D., Lu Z., Schechter A., Gofer Y., Gizbar H., Turgeman R., Cohen Y., Moshkovich M., Levi E. Prototype systems for rechargeable magnesium batteries. *Nature* **2000**, 407, 724-727.
- [2] Li C., Guha R. D., Shyamsunder A., Persson K. A.\*, Nazar L. F.\* A weakly ion pairing electrolyte designed for high voltage magnesium batteries. *Energy & Environ. Sci.* **2024**, 17, 190-201.

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